World-Model Value Depends on the Policy That Acts

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Abstract: We argue that the value of a world model for robot control should be evaluated jointly with the policy that acts on its predictions. A policy's ability to perform a task and its ability to benefit from prediction need not improve in the same way. In a simulated manipulation task, we hold predictions and the actions selected from them fixed while varying the policy that subsequently controls the robot. We find that improved baseline performance can accompany greater benefit from the same selected actions. Comparisons with action selection based on current observations further distinguish the benefit of an intervention from the added value of future prediction. These findings motivate evaluating world models across acting policies, with explicit comparisons that establish when predictive information improves control.

Keywords: World Models, Robot Learning, Decision Value, Evaluation

1 Introduction

A world model contributes to control when its predictions lead to better actions. Policy improvement changes both performance without prediction and the consequences of using it. We argue that **world models should be evaluated jointly with the acting policy, the way predictions influence its actions, and the time available to act.**

Figure 1 illustrates how advance information can enable movement toward a predicted target, while effective feedback can reduce its benefit. These hypothetical capabilities have no prescribed learning order.

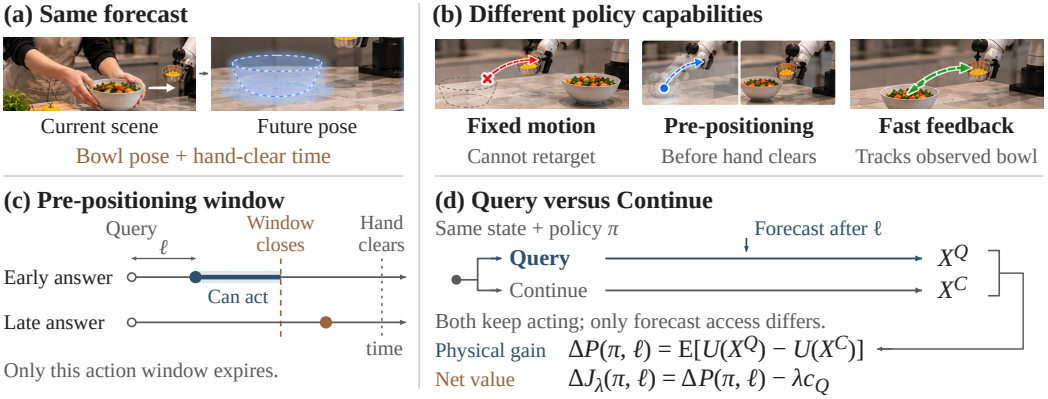


Figure 1: **The same forecast can support different actions.** Conceptual illustration of policy capabilities and an idealized physical-time comparison in panel (d). U denotes task utility, $\ell = \tau$ is prediction delay, and c_Q is additional query computation cost, weighted by λ. Delay losses are reflected in the trajectories and excluded from λc_Q. Our experiments pause physics during inference; Section 3.2 fixes the selected actions across policies.

10th Conference on Robot Learning (CoRL 2026), Austin TX, USA.

Value of information and metareasoning connect information and computation to the decisions they improve [1, 2]. Value-aware model learning, value equivalence, and objective mismatch similarly motivate evaluating predictions through their consequences for control [3, 4, 5].

We study a short sequence of selected commands, termed *action advice*, after which a *continuation policy* resumes control from ordinary observations. Our evaluations distinguish the value of identical advice across policies from the additional benefit of predictive features.

2 Decision Value of World-Model Predictions

2.1 Policy-dependent decision value

Let W be a fixed predictor, π an acting policy, I the mechanism that converts predictions into actions, and τ the prediction delay. For matched initial conditions, define

$$V(W, \pi, I, \tau) = \mathbb{E}[J(\mathrm{query}(W, \pi, I, \tau)) - J(\mathrm{continue}(\pi))] \,. \tag{1}$$

Here, query denotes execution with prediction-based intervention, whereas $\mathrm{continue}$ uses the policy alone. The utility J specifies task performance and any penalties for time, risk, or computation.

Our empirical measure focuses on task performance: $G = \mathbb{E}[D_{\mathrm{base}} - D_{\mathrm{query}}]$, where D is mean object–goal distance over steps 91–100 of a 100-step episode. This final window measures performance after the policy has resumed control and reduces dependence on a single terminal sample. Positive G indicates reduced distance. Computation and physical delay are separate considerations in the broader utility J.

Policy ability. Baseline performance and intervention benefit measure different properties. Success, tracking, recovery, and action latency also capture distinct capabilities; training updates identify saved policies rather than a calibrated competence scale. We therefore report baseline performance and intervention gain separately across all evaluated checkpoints.

2.2 Controlled and natural evaluation

In *fixed-advice evaluation*, we restore the same simulator state for two executions. One executes five baseline commands and the other executes five selected commands; the same policy π then completes both executions. We call the shared starting state a *fork* and the initial five-command sequence an action *prefix*.

Within each fork, the initial observations, forecasts, candidate sequences, and selected prefix k are fixed across continuation policies. For final-window error $D(\pi, k)$, the intervention gain is $G(\pi; k) = \mathbb{E}[D(\pi, 0) - D(\pi, k)]$, where candidate 0 supplies the common baseline prefix. Reporting this gain alongside $\mathbb{E}[D(\pi, 0)]$ separates the value of the intervention from the performance of the corresponding baseline execution.

In *natural-execution evaluation*, each policy generates candidates at the states it visits, and action selection is applied repeatedly. This evaluates the complete intervention procedure relative to execution by that policy alone.

Prediction-based and direct scoring. The frozen world model (WM) predicts visual and proprioceptive features for each candidate sequence. A learned scorer uses these features to estimate candidate outcomes. A *direct scorer* instead uses current features, with the same pretrained encoder, input types, candidate construction, network capacity, and simulated-outcome supervision. Each scorer has its own validation-selected intervention threshold. Their comparison therefore measures the added benefit of predicted features within the resulting selection procedures. A shuffled-forecast control additionally tests whether forecasts must correspond to the actions being evaluated.

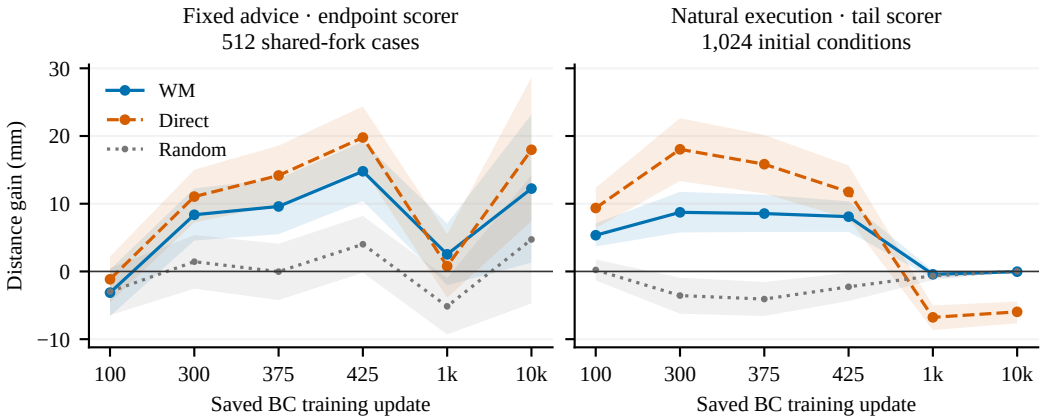


Figure 2: **Policy-dependent benefit under two interventions.** Left: identical advice selected using predicted endpoint distance, followed by each continuation policy (512 shared states). Right: repeated selection using predicted final-window distance during each policy's execution (1,024 cases). Positive gain reduces final-window distance. Shading shows descriptive paired 95% intervals; formal tests use a prespecified 15-comparison family. Scoring targets, thresholds, intervention schedules, and baselines differ between panels.

3 Experiments

3.1 Experimental setup

We use a fixed pretrained JEPA-WM [6] and behavior-cloning (BC) policies on the Meta-World `push-v3` task [7]. The policies observe simulator state. BCu denotes the policy saved after u training updates; we evaluate all six checkpoints at 100, 300, 375, 425, 1,000, and 10,000 updates from one training run. The 100-to-300 comparison was selected during development and specified before evaluation on fresh episodes.

Scorers rank nine candidate sequences of 20 actions. Only the first five actions execute before BC resumes control. The fixed-advice scorer is trained to predict distance after 20 actions without feedback. The natural-execution scorer is trained to predict final-window distance after five selected actions and subsequent BC control.

The primary evaluations share 15 prespecified directional tests with total $\alpha = .010$. We use Bonferroni-adjusted one-sided bootstrap lower bounds; a positive bound supports the specified direction of an effect. Figure 2 shows all checkpoints, and Appendix A provides training and statistical details.

3.2 Fixed-advice evaluation

BC300 generates shared states and candidates at step 10 in 512 cases. Each baseline or selected prefix executes to step 15, after which the tested continuation policy controls the robot to step 100 without further advice.

Replacing BC100 with BC300 increases WM intervention gain from -3.129 to 8.369 mm, a rise of 11.498 mm, while baseline error decreases by 22.434 mm. Gain at BC300, the increase in gain, and baseline improvement all have positive family-adjusted lower bounds (Table 1). Improved baseline performance therefore accompanies greater benefit from identical advice. Direct scoring also exhibits increasing gain: the WM-minus-direct difference in increases is -0.716 mm, with a family-adjusted lower bound of -7.141 mm.

3.3 Natural-execution evaluation

We evaluate repeated intervention on 1,024 cases, selecting actions at steps $5, 10, \ldots, 80$. The baseline, *Continue*, uses BC throughout and generates a new action sequence every five steps.

At BC300, WM selection improves final-window distance by 8.729 mm, compared with 18.036 mm for direct scoring. The paired WM–direct difference is -9.307 mm. Both procedures improve on Continue, with the larger gain obtained by direct scoring. WM gain increases by 3.379 mm from BC100 to BC300, but its family-adjusted lower bound of -2.445 mm leaves this increase statistically unresolved. The decreases from BC300 to BC1000 and BC10000 have positive family-adjusted lower bounds.

Appendix B extends these comparisons to scorers that also receive a representation of the continuation policy.

4 Discussion

Task objectives and remaining improvement. Continuous distance and changes in success complement binary task completion. Precision may remain important near success saturation, while disturbance recovery requires separate evaluation. A post-hoc diagnostic illustrates why the evaluation horizon also matters: at BC1000, a selector given the actual simulated outcomes of all candidate plans improves 20-step endpoint distance by 7.790 mm but worsens final-window distance by 3.677 mm relative to the same-duration baseline plan (Appendix C). Optimizing a short-horizon objective can therefore conflict with subsequent policy control.

Inference time and physical delay. With physics paused, natural-execution candidate evaluation at BC300 takes 1.001 s per episode for WM versus $.057$ s for direct, including rendering and encoding. Measuring the task cost of waiting requires continuous physical execution, a specified controller during inference, and equal-duration comparisons. Task gains and computation should therefore be reported separately.

Online and offline computation. World models can support action selection during execution or contribute to policy learning. Planning at these stages has distinct effects [8]; guided policy search, for example, uses trajectory optimization to guide policy learning [9]. Our experiments concern online action selection. Extending policy-dependent evaluation to model-assisted learning is a complementary direction.

Limitations. Our findings are specific to one task, one BC training seed, privileged observations, and the chosen candidates, representations, and scorers. They establish an instance of policy-dependent intervention value, without isolating individual capabilities or a general competence–benefit relationship. The context study is exploratory and reuses evaluated cases. Generalization to vision–language–action policies, disturbance recovery, and real-time control remains open.

5 Conclusion

We advocate evaluating world models jointly with the policies that act on their predictions. Our controlled experiment shows that identical prediction-based advice can become more valuable as baseline performance improves. Direct-scoring comparisons further show why intervention benefit and predictive advantage require separate evaluation. Benchmarks should therefore report gains across acting policies, include controls without predicted features, and distinguish task performance from inference cost. This policy-dependent perspective makes the conditions under which predictions improve control central to world-model evaluation.

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A Experimental Details and Primary Results

All distances are in millimeters unless stated otherwise.

A.1 Environment, policies, and prediction interface

We use Meta-World `push-v3` with $0.0125\,\mathrm{s}$ native steps and 100-step evaluation episodes. BC observes 39 simulator-state coordinates plus the expert-goal hand position and gripper state. It outputs 20 four-dimensional actions; five execute before a new chunk is generated. Its two 256-unit ReLU hidden layers and tanh output are trained with masked mean-squared error, Adam (3×10^{-4}), batch size 256, and seed 71. The six checkpoints come from this one 10,000-update run. An input correction clamps coordinates constant in the BC training data to their training values; weights and stored normalizers are unchanged.

The fixed public Meta-World JEPA-WM has a DINOv2 ViT-S/14 encoder and six-layer predictor [6]. A 224×224 RGB view and proprioception produce 6,144 visual coordinates (tokens pooled to 4×4) and 4,096 proprioceptive coordinates. Scorers receive current/goal representations, time $t/100$, base and candidate actions, and either candidate-conditioned future features or, for direct scoring, repeated current features. The encoder and predictor are never fine-tuned. Direct scoring retains the shared pretrained encoder.

Candidates comprise the BC proposal and four pairs of opposite perturbations: translational standard deviation .15, constant within each five-step block, unchanged gripper, and clipping to $[-1, 1]$. Episode/decision/pair-keyed perturbations are shared across methods. Candidate actions can nevertheless differ when the underlying BC proposals differ.

A.2 Scorer training and selection

Scorers are fitted on 256 episodes and validated on 32 separate episodes. Each supplies nine forks: BC300/425/1000 at steps 10/30/50, with nine candidate branches per fork. Endpoint labels measure distance after 20 open-loop actions; tail labels measure final-window distance after five actions and policy continuation. Both primary evaluations forecast 20 steps ahead but execute only five candidate actions.

Each scorer is a multilayer perceptron with 256/128 ReLU hidden units and one distance output. Training uses 20 epochs, Adam (learning rate .001, weight decay .0001), batches of 32 decision groups, gradient clipping at 1, and seed 71. Inputs use training-only means/scales computed in float64, with unit scale for constant coordinates. The loss combines smooth-L_1 regression ($\beta = .02\,\mathrm{m}$) and pairwise ranking (weight .005, temperature $.005\,\mathrm{m}$), excluding pairs separated by at most $.001\,\mathrm{m}$. Epoch selection minimizes validation tail regret (excess tail distance over the best candidate), including for endpoint-trained heads; ties favor the earlier epoch. Each head's margin in $\{0, .005, .010\}\,\mathrm{m}$ maximizes validation gain over natural Continue at BC300; ties favor the larger margin.

Given scores s_k, the selector chooses $k^* = \arg\min_k s_k$, breaking ties by smallest index, and overrides candidate 0 only when $s_0 - s_{k^*} > m$. WM/direct margins are $.005/0\,\mathrm{m}$ in fixed-advice evaluation and $.010/0\,\mathrm{m}$ in natural execution. Thresholds are selected separately for each scorer. The gated-random baseline shares the WM gate but samples a seeded random nonbase candidate, providing an action-selection control distinct from shuffled forecasts.

A.3 Paired evaluation and statistical protocol

Natural-execution evaluation uses the first 1,024 eligible episodes from a fresh reserved pool; fixed-advice evaluation uses their first 512, selected before outcomes. Eligibility requires ten consecutive expert successes within 500 steps, with at least 100 expert steps; three of 1,027 attempts failed this criterion. Goal RGB/proprioception is cached at the first qualifying window's end and shared across conditions. The protocol disallows replacement after technical failure.

Natural execution applies selection at steps $5, 10, \ldots, 80$ along each policy's trajectory. Continue uses the same BC weights, initial state, goal, action limits, five-step refresh, and horizon without querying. Fixed-advice evaluation instead uses BC300 to reach step 10 and generate nine cached candidates. Each selector's chosen prefix executes to step 15, after which the tested continuation controls the robot to step 100. The common candidate-0 baseline is therefore hybrid for policies other than BC300. On the shared 512 cases, its BC300 traces exactly match natural Continue300. Each continuation determines subsequent feedback and visited states. Physics is paused during inference in both settings.

The primary outcome averages object–goal distance over steps 91–100 of the same rollout, summarizing final-window performance rather than immediate prefix progress. Terminal success corresponds to final distance at most 50 mm. A rescue denotes a failed baseline rollout paired with a successful intervention rollout from the same initial condition; a harm reverses these outcomes. Both primary evaluations share 15 prespecified directional contrasts with total $\alpha = .010$. Family lower bounds are the $.010/15$ quantile of 20,000 paired episode bootstrap replicates (seed 5263), retaining each study's denominator. These approximate bounds condition on the trained models and task. Bonferroni allocation accounts for the shared family across both studies. Ordinary paired 95% intervals are descriptive. The fixed-advice claim requires positive family bounds for BC300 gain, its increase over BC100, and baseline improvement jointly. All tests retain their prespecified direction.

Selection procedures and the 100-to-300 comparison were chosen during development, then fixed with contrasts and sample sizes before evaluating the fresh pool. We report all six checkpoints. Earlier endpoint, label-efficiency, and delayed-advice probes did not support their primary claims. The continuation-aware study followed the primary evaluations and is exploratory.

Table 1: All 15 prespecified contrasts, in millimeters. A positive family lower bound supports the corresponding directional claim under the stated bootstrap procedure. The 95% column gives ordinary paired intervals. Baseline improvement means a decrease in baseline error.

Study	Contrast	Mean	95% interval	Family lower
Natural	WM gain at 300	8.729	[5.753, 11.760]	3.835
Natural	WM − direct at 300	-9.307	[-13.750, -4.767]	-16.865
Natural	WM − random at 300	12.291	[8.963, 15.726]	6.706
Natural	WM gain: 300 − 100	3.379	[0.042, 6.815]	-2.445
Natural	WM gain: 300 − 1,000	9.155	[6.123, 12.256]	4.113
Natural	WM gain: 300 − 10,000	8.762	[5.778, 11.798]	3.864
Natural	Baseline improvement: 100 to 300	70.902	[66.041, 75.724]	63.168
Natural	Baseline improvement: 300 to 1,000	96.318	[92.633, 99.944]	90.409
Natural	Baseline improvement: 300 to 10,000	98.269	[94.750, 101.744]	92.579
Natural	WM–direct gain: 300 − 100	-5.290	[-10.510, -0.047]	-14.041
Fixed advice	WM gain at 300	8.369	[4.561, 12.266]	2.095
Fixed advice	WM gain: 300 − 100	11.498	[6.803, 16.249]	3.920
Fixed advice	Baseline improvement: 100 to 300	22.434	[16.783, 28.106]	13.385
Fixed advice	WM rise − direct rise	-0.716	[-4.581, 3.048]	-7.141
Fixed advice	WM rise − random rise	7.082	[1.859, 12.404]	-1.754

In natural execution at BC300, success rates are 14.160%/20.313%/35.352% for Continue/WM/direct; WM and direct produce 82/19 and 271/54 rescues/harms, respectively. Candidate evaluation takes 1.00063 s for WM and .05700 s for direct per episode, including rendering, encoding, prediction, scoring, and transfers. These pipeline times include queries that retain candidate 0, not just overrides, and do not measure physical waiting loss.

Table 2: All checkpoint means in millimeters. Lower baseline error and higher gain are better; gains are relative to each row's baseline. The oracle uses outcomes to choose the best of nine prefixes at each fixed fork. It measures candidate-set opportunity and is not deployable. No oracle was collected for natural-execution evaluation.

Study	Update	Base error	WM gain	Direct gain	Random gain	Oracle gain
Fixed advice	100	143.514	-3.129	-1.160	-2.975	36.291
Fixed advice	300	121.080	8.369	11.054	1.441	46.420
Fixed advice	375	115.361	9.596	14.171	-0.030	49.076
Fixed advice	425	123.299	14.788	19.777	4.017	54.410
Fixed advice	1,000	66.123	2.530	0.784	-5.154	28.613
Fixed advice	10,000	109.609	12.236	17.964	4.725	76.718
Natural	100	189.751	5.350	9.367	0.224	—
Natural	300	118.849	8.729	18.036	-3.562	—
Natural	375	85.844	8.551	15.837	-4.080	—
Natural	425	62.469	8.089	11.746	-2.268	—
Natural	1,000	22.530	-0.426	-6.776	-0.641	—
Natural	10,000	20.579	-0.034	-5.965	0.031	—

B Continuation-Aware Action Selection

B.1 Design and controls

This exploratory study tests whether continuation context improves action selection with forecasts beyond direct scoring. It reuses all 512 states from the fixed-advice evaluation; natural execution uses the first 256, selected by a runtime rule before inspecting the follow-up results. We report ordinary paired 95% bootstrap intervals (20,000 replicates, seed 5263).

Training and validation use the same 256/32 episodes and nine saved-state types as the primary studies, crossing nine prefixes with six continuations to obtain tail labels. Fitting, epoch selection, and gating use labels only for BC100/300/1000/10000. BC375/425 labels are withheld from fitting, but BC425 contributes training trajectories and both checkpoints were examined during development. Continuation context is a 160-dimensional fingerprint formed by concatenating each policy's five four-dimensional actions at eight fixed probe states: BC300's step-10 states in the first eight training episodes, with cached goals. The fingerprint contains no outcome labels and is constant across candidates; plain scorers receive 160 zeros.

All seven heads share the augmented input width, architecture, loss, optimizer and fitting budget above. Normalization uses training inputs and fitting-context fingerprints. Epoch selection minimizes mean ungated regret across fitting continuations. The same margin grid maximizes cached-branch validation gain over candidate 0, with larger-margin ties. All six deployable heads choose zero; the privileged actual/context head chooses .005 m.

WM features describe the five-action prefix endpoint; the implementation unrolls all four blocks and reads the first prediction, saving no predictor computation. Shuffling permutes nine forecasts with no fixed points, preserving current/goal/action inputs. Its episode/time-keyed permutation is shared across policies and applied during both fitting and evaluation. Shuffling tests action–forecast correspondence; direct scoring tests predictive advantage. The separately trained actual/context head uses the encoded physical observation after the same prefix; it is unavailable at deployment and is not a guaranteed upper bound. Relative to the primary evaluation, this design changes the context inputs, supervision, and feature horizon together.

The controlled evaluation crosses six contexts with six continuations. Policy-specific context can change the advice; fixing BC300's fingerprint preserves identical advice within each fork. For every continuation, a different policy's fingerprint yields the largest mean benefit, indicating a mismatch between preferred context and acting policy. Tables 3–4 report all contrasts and curves.

Table 3: Exploratory continuation-context contrasts on 512 shared states, in millimeters. The first eight rows use continuation 300 and context 300 where applicable; context benefit is context minus plain. The last two rows compare continuations 100 and 300 while always supplying context 300. The first row is the primary comparison of WM and direct scoring.

Contrast	Mean	95% interval
WM/context − direct/context at 300	1.951	[−.955, 4.884]
WM/context gain at 300	17.081	[12.505, 21.777]
WM context benefit	5.831	[2.270, 9.469]
Direct context benefit	4.186	[1.400, 7.008]
WM context benefit − direct context benefit	1.646	[−2.455, 5.762]
Aligned − shuffled, with context	11.602	[7.371, 15.900]
Aligned − shuffled, without context	6.408	[1.999, 10.849]
Actual/context − WM/context	−9.022	[−13.135, −4.959]
BC300 context fixed: WM rise, 100 to 300	6.970	[1.536, 12.518]
BC300 context fixed: WM rise − direct rise	4.359	[.631, 8.156]

Table 4: All exploratory mean gains (millimeters) on 512 shared states, with policy context where applicable. The baseline and oracle are those in the fixed-advice rows of Table 2.

Head	100	300	375	425	1,000	10,000
Direct/plain	9.183	10.945	12.199	15.046	2.761	.241
WM/plain	8.297	11.250	10.409	14.581	6.735	2.280
Direct/context	12.008	15.131	12.856	16.906	3.601	3.455
WM/context	6.362	17.081	18.064	20.367	5.478	11.737
Shuffled/plain	1.401	4.842	3.965	6.932	−1.075	1.385
Shuffled/context	2.254	5.480	4.149	5.792	−.100	2.173
Actual/context	5.481	8.059	10.480	12.331	5.486	7.768

B.2 Matched natural execution

For natural execution, the six deployable frozen scorers select at steps $5, 10, \ldots, 80$, executing five actions per selection; context scorers receive the acting policy's fingerprint. Continue refreshes BC without querying, and the privileged actual/context scorer is excluded. Evaluation comprises all $256 \times 6 \times 7$ trajectories, including Continue. Cases are reused from the primary evaluation. Tables 5–6 report matched comparisons and all natural-execution checkpoints.

Table 5: Continuation-aware selection on the same 256 cases: means and ordinary paired 95% intervals, in millimeters. Both columns use each continuation's own context where applicable. "Context benefit" means context minus plain; the fifth row subtracts the direct context benefit from the WM context benefit. Intervals describe paired method comparisons within each evaluation setting.

Contrast at BC300	Controlled fork	Natural execution
WM/context gain	15.618 [9.016, 22.564]	15.232 [5.735, 24.790]
WM/context − direct/context	0.874 [-3.116, 5.052]	-12.600 [-21.737, -3.668]
WM context benefit	4.451 [-0.274, 9.530]	-2.660 [-11.526, 5.972]
Direct context benefit	3.553 [-0.377, 7.612]	13.631 [5.983, 21.308]
WM − direct context benefit	0.899 [-4.664, 6.592]	-16.291 [-28.006, -4.619]
Aligned − shuffled, with context	9.608 [3.440, 15.963]	9.017 [-0.595, 18.836]

At BC300, both WM and direct improve on Continue, with the larger gain obtained by direct scoring (Table 5). The natural-execution WM rise is uncertain at 3.446 mm $[-7.827, 14.560]$. Controlled/natural alignment estimates are 9.608/9.017 mm, with different uncertainty. The own-context comparison permits advice to change; fixing context to BC300 gives an identical-advice rise of 5.926 mm $[-1.413, 13.429]$ on these 256 cases. The two settings jointly differ in visited states, candidates, and intervention frequency.

Table 6: Complete continuation-aware natural-execution means ($N = 256$), in millimeters. The first row is baseline error; every remaining row is gain over that baseline. Positive gain reduces distance; negative gain increases it.

Method	100	300	375	425	1000	10000
Continue error	194.840	120.771	84.280	67.207	22.983	21.190
Direct/plain	7.626	14.201	6.822	9.448	-6.343	-5.807
WM/plain	13.242	17.892	8.441	1.272	-18.349	-21.047
Direct/context	9.436	27.831	21.043	19.618	-3.216	-0.984
WM/context	11.786	15.232	8.463	4.957	-19.003	-11.015
Shuffled/plain	5.353	9.351	3.334	1.993	-4.904	-1.604
Shuffled/context	5.342	6.215	2.346	5.981	-2.897	-1.483

Natural-execution success at BC300 is 35/256, 63/256, and 98/256 for Continue, WM/context, and direct/context. WM produces 51 rescues and 23 harms; direct produces 72 and 9. Candidate-evaluation times are .9994/.0571 s per episode for WM/direct under paused physics.

C Short-Horizon Optimization and Final-Task Performance

This post-hoc analysis uses 256 separate eligible episodes and all six checkpoints, with the original BC input interface rather than the constant-coordinate correction in Appendix A. Episodes are disjoint from the primary evaluations; the policy training seed is unchanged. At step 30, a privileged oracle selects among nine 20-action plans using their *actual simulated* endpoint object–goal distances. The selected plan executes to step 50, followed by BC control to step 100. Base20 commits the unmodified candidate for 20 steps, whereas Continue retains five-step feedback. Physics is paused during computation. The oracle evaluates the physical endpoint objective using simulated outcomes.

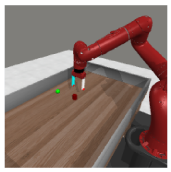
Table 7: Physical-endpoint oracle: all checkpoints and both references, with ordinary paired 95% intervals ($N = 256$, post hoc; millimeters). Positive gain reduces the corresponding physical distance. Base20 commits candidate 0 for 20 actions; Continue retains five-step feedback.

Update	Reference	Endpoint gain	Final-task gain
100	Base20	5.602 [4.521, 6.757]	5.787 [3.514, 8.294]
100	Continue	4.820 [3.770, 5.917]	6.977 [4.365, 9.780]
300	Base20	11.549 [10.211, 13.039]	15.295 [11.246, 19.480]
300	Continue	10.851 [9.205, 12.676]	16.044 [11.285, 20.932]
375	Base20	13.037 [11.877, 14.248]	16.210 [11.147, 21.429]
375	Continue	11.465 [10.008, 12.982]	21.353 [15.921, 26.894]
425	Base20	11.676 [10.658, 12.783]	11.159 [6.181, 16.456]
425	Continue	8.437 [7.157, 9.775]	14.181 [9.222, 19.323]
1000	Base20	7.790 [7.145, 8.478]	-3.677 [-6.150, -1.737]
1000	Continue	4.879 [4.191, 5.592]	-3.761 [-6.354, -1.675]
10000	Base20	5.100 [4.602, 5.636]	-0.698 [-2.126, 0.444]
10000	Continue	4.456 [3.946, 5.005]	-0.714 [-2.151, 0.441]

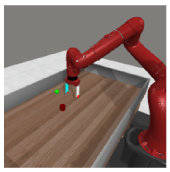
BC1000 provides a post-hoc example of opposing endpoint and final-task effects: endpoint distance improves relative to Base20 by 7.790 mm [7.145,8.478], but its final gain is -3.677 mm $[-6.150, -1.737]$. Because candidate 0 is included among the choices, nonnegative endpoint gain against Base20 is expected by construction. The final-task loss shows that improving the physical endpoint objective can worsen performance under policy continuation.

D Physical Replay Illustrations

Figure 3 illustrates the continuation-aware experiment in Appendix B. For each case, BC300 reaches the shared state at step 10. The robot executes either baseline candidate 0 or selected candidate 3 for five steps, then BC100 or BC300 controls it to step 100. The selector receives BC300's policy representation in both comparisons, so the selected advice is identical across continuations.



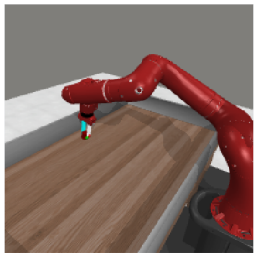
Case (a): episode 880102
Shared state at step 10



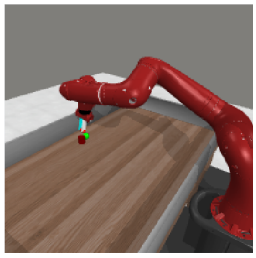
Case (b): episode 880316
Shared state at step 10

BC100: baseline | BC100: advice | BC300: baseline | BC300: advice

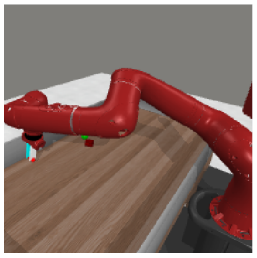
(a) Increasing intervention gain: $-36.3 \rightarrow +2.6$ mm



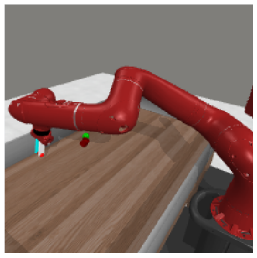
$D = 9.6$ mm



$D = 45.9$ mm

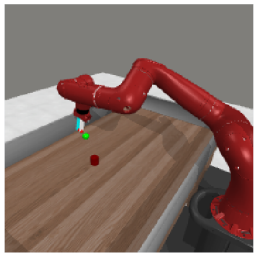


$D = 82.4$ mm

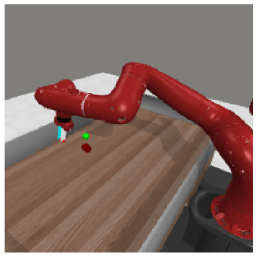


$D = 79.7$ mm

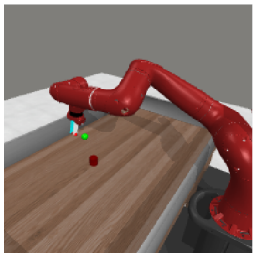
(b) Decreasing intervention gain: $+116.0 \rightarrow +71.8$ mm



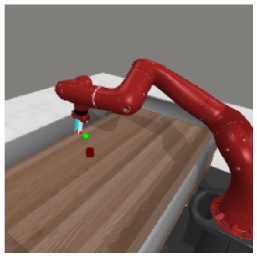
$D = 243.3$ mm



$D = 127.3$ mm



$D = 243.3$ mm



$D = 171.5$ mm

Figure 3: **Identical advice can have different effects under different continuation policies.** The upper thumbnails show the shared starting states; the eight larger panels show actual simulated states at step 100. Labels give D, mean object–goal distance over steps 91–100, rather than distance in the displayed final frame. Gain is baseline D minus advice D. These are physical replays from the exploratory continuation-aware study, not decoded predictions or episodes selected to represent the primary result in Figure 2.

Cases were selected after evaluation by taking the episode nearest the median gain change within each sign group, with ties resolved by episode ID. Among 512 cases, 252 changes were positive, 177 negative, and 83 zero. In case (a), baseline error increases from 9.6 to 82.4 mm: the increased intervention gain mainly reflects reduced harm, not improved baseline performance. Case (b) shows reduced benefit from the same advice. These examples illustrate variation across cases; the population-level claim relies on the complete paired evaluations.